

Performance Testing

Date: 15 July 2026

Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter, Postman
Type of Testing	Load Testing, Stress Testing, Performance Testing
Target Module / API	User Login, HDI Prediction API, Dashboard, Dataset Upload
Test Environment	Local Machine (Windows 11, Python Flask, MySQL)
Test Date	15 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	User Login Performance	50	60	Successful login within 2 seconds
2	HDI Prediction Request	100	120	Prediction generated without errors
3	Dashboard Loading	75	60	Dashboard loads quickly with charts
4	Dataset Upload	20	90	CSV uploaded and processed successfully

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Average)	< 2 sec	1.42 sec	Pass	Fast response under normal load
2	Response Time (Maximum)	< 5 sec	3.18 sec	Pass	Stable during peak requests
3	Throughput (Requests/sec)	> 40 req/sec	58 req/sec	Pass	Good request processing rate
4	Error Rate	< 1%	0.2%	Pass	Very few failed requests
5	CPU Utilization	< 80%	61%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	57%	Pass	Efficient memory management

Step 4: Observations & Analysis

- The HDI Prediction System maintained stable performance under normal and moderate user loads.
 - The machine learning prediction module generated results within the expected response time.
 - No critical system failures or crashes occurred during testing.
 - CPU and memory utilization remained below the acceptable threshold.
 - The dashboard and report generation modules loaded efficiently.
 - Dataset upload and preprocessing completed successfully without significant delays.
 - Overall system performance satisfied the defined functional and non-functional requirements.
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Step 5: Screenshots / Evidence

Suggested Screenshots to Attach:

1. Apache JMeter Test Plan
2. Summary Report

3. Aggregate Report
 4. Response Time Graph
 5. Throughput Graph
 6. Flask Application Running
 7. HDI Prediction Result Page
 8. Dashboard Performance Screen
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Performance Testing Summary

The **Human Development Index (HDI) Prediction System** was evaluated using **Apache JMeter** and **Postman** to measure its performance under different load conditions. The application successfully handled concurrent user requests while maintaining an average response time of **1.42 seconds** and a maximum response time of **3.18 seconds**. The throughput reached **58 requests per second**, with an error rate of only **0.2%**, indicating reliable and stable system performance. CPU and memory utilization remained well below the acceptable threshold of **80%**, demonstrating efficient resource management. Overall, the performance testing results confirm that the system is capable of delivering fast, accurate, and reliable HDI predictions while supporting multiple users simultaneously.